

Conceptual Data Architecture for a Statistical Office

A sovereign, on-premise solution for Centrala Statistikas Parvalde data ecosystem, from sources to publication.

PostgreSQL + R · CSP's own core

Data never leaves Latvia

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Domain for the data model:

Population & Demography

One database, three zones, computation in R

Four sources are collected into **one PostgreSQL database**, split into three readiness zones (raw, clean, mart). Indicators are computed in **R**. Before release the data passes a mandatory disclosure-risk check, after which it is published to an open data portal and to Eurostat.



Fit the task, not the brochure

Four sources, one domain, national-register volumes. A task this size does not need a large platform. The right answer is an architecture that is only as big as it needs to be.



Build on what CSP already runs

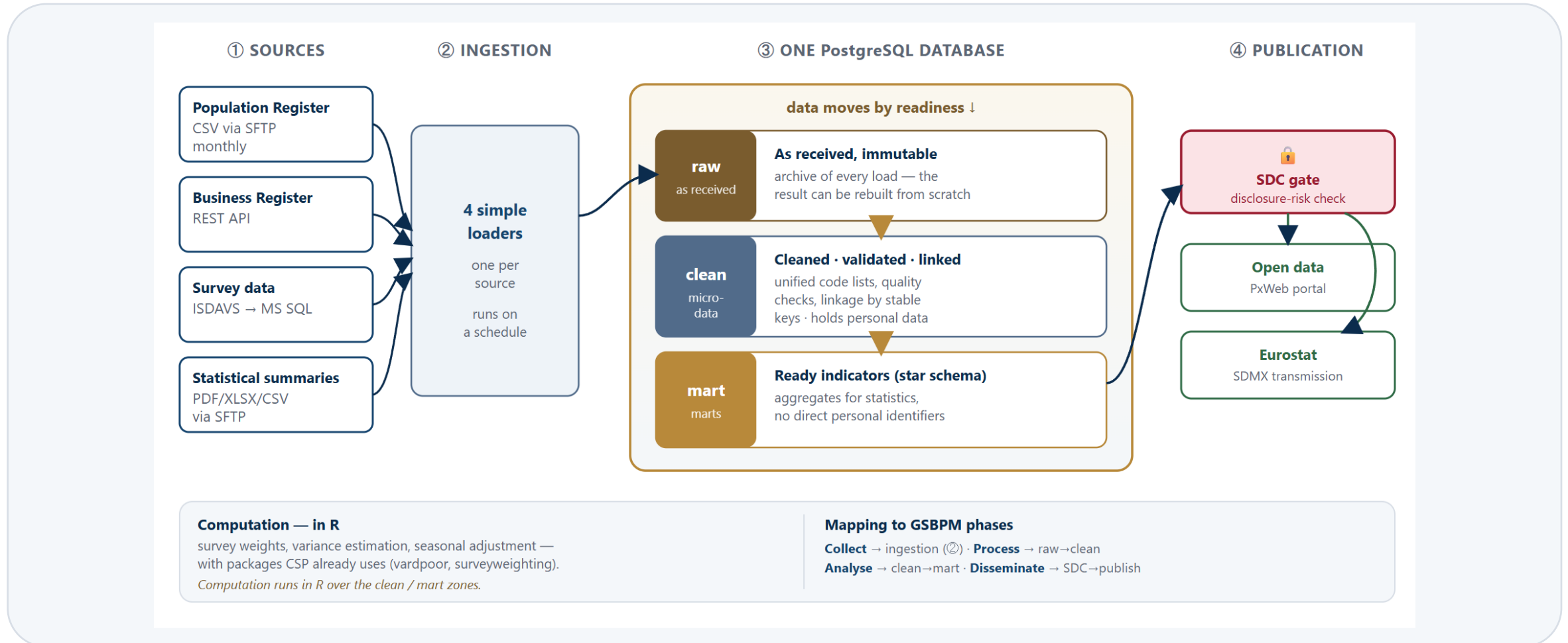
CSP's public code shows a **PostgreSQL + R** stack. This design extends it rather than replacing it.

Four sources, four transports

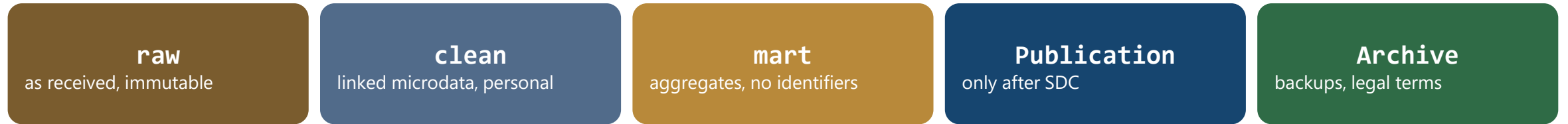
Source	How it arrives	What it brings
Population Register	Monthly CSV over SFTP Tools: an SFTP client such as paramiko or pysftp in Python or plain ssh/curl.	The frame for demography: who is resident, by age, sex, territory, citizenship
Business Register	REST API Tools: http2 in R or requests/httpx in Python.	The enterprise frame, linked to people through stable identifiers
Survey data (ISDAVS)	MS SQL relational database Read from a read-replica of the survey database (a copy) or through SQL Server's Change Tracking / CDC	Sample surveys; the existing Microsoft footprint stays a source
Statistical summaries	SFTP: PDF / XLSX / CSV CSV and Excel are read directly with readxl/readr in R or openpyxl/pandas in Python. PDF is the awkward one: tables are pulled out with camelot or tabula	Pre-made figures; the one fragile source, handled with care

The job: turn these four very different feeds into **trustworthy official statistics**, published to Eurostat and to the public, without ever exposing a single identifiable person.

From sources to publication, in a straight line



Three zones by readiness, one hard line



Why three zones

raw is append-only, so any figure can be rebuilt from its input.
clean is the single trusted layer for every calculation.
mart holds only aggregates, ready to publish.

The defining control

Identifiable microdata (raw, clean) is kept apart from aggregates, by storage and by permissions. **No single role goes from ingestion to publication.** The only way out is the SDC gate.

Confidentiality is built in, not bolted on



What the gate does

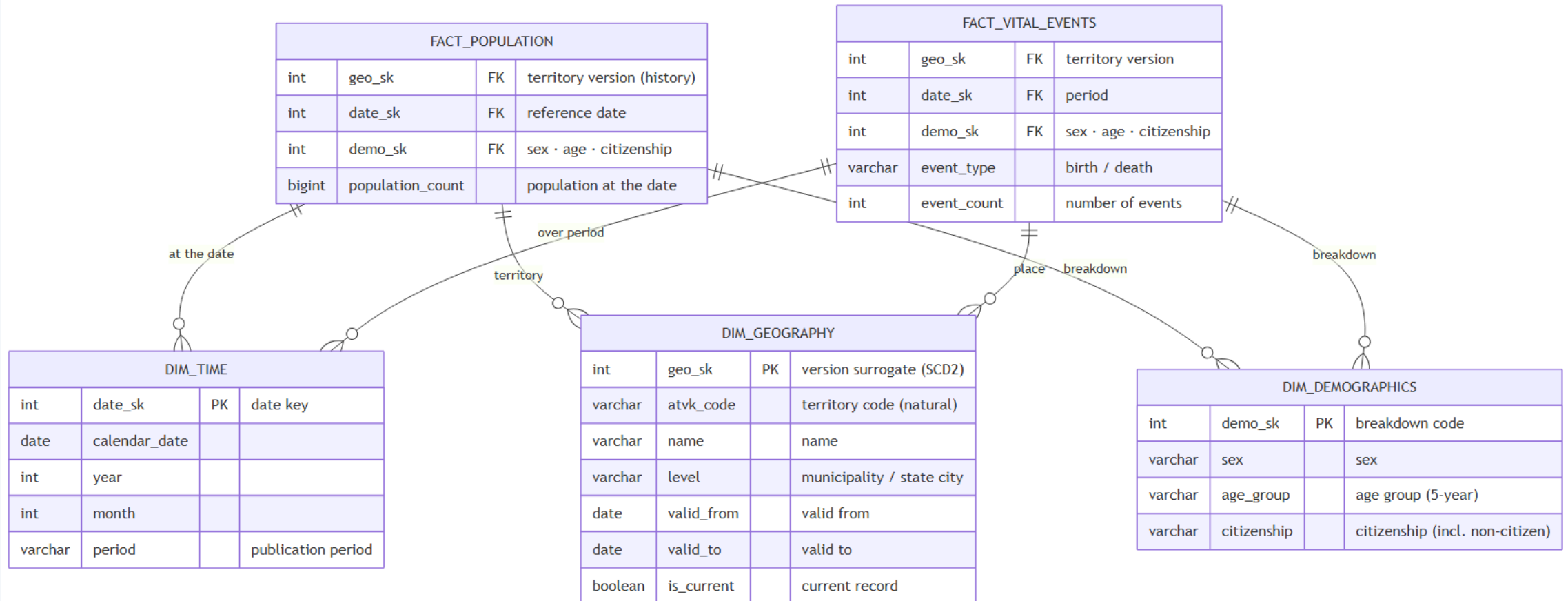
Suppresses or rounds small cells and checks that a new release does not, by subtraction, reveal what an earlier one hid. In R this is the open, Eurostat-maintained `sdcTools` (with tau-ARGUS for tables).

Why it is enough here

One mandatory step on the only exit, a record of what was suppressed and clearance kept separate from production. Simple, auditable and hard to bypass.

Role	raw	clean (microdata)	mart (aggregates)	SDC gate	Publication	Access log
Loaders (service accounts)	write	—	—	—	—	writes events
Methodologist / statistician	read (by task)	read/write	read/write	submit	—	—
Disclosure officer (SDC)	—	read (risk)	read	clearance	—	read (own actions)
Publication analyst	—	—	read (cleared only)	—	publish	—
Administrator / data architect	read	read (audit)	read	operate	—	read + config

Population & Demography as a star schema



A series that breaks if you ignore history

119

units before 2021
(110 + 9 cities)



43

after the 2021 reform
(36 + 7 state cities)



42

from 1 July 2025
(35 + 7 state cities)

When **Varaklani** merged into **Madona** on 1 July 2025, the map changed again. If `dim_geography` did not keep the validity period of each code, the series for Varaklani and Madona before and after that date would not line up. The year-on-year comparison would simply be wrong.

The fix is one design decision: facts point to the *version* of a territory that was valid on the reference date, not to the territory in general. That keeps every time series comparable through any reform.

The steps already map to the standards



GSBPM

The common map of statistical-production phases. Our four steps map straight onto it.

SDMX

The standard transmission format to Eurostat. Here it is simply the output of the mart zone.

SDC

The confidentiality check before publication, the one statistical control that cannot be skipped.

Metadata runs across every step: indicator definitions, classifications and lineage. Demography rests on **Regulation (EU) No 1260/2013**, moving to **2025/2458** from 2028.

Small, sovereign and auditable

Strengths

- **Sovereignty.** Everything stays local. Data never leaves Latvia.
- **Confidentiality.** Microdata and aggregates are separated, with a mandatory SDC gate.
- **Integration by stable keys,** with quality control and territory history.
- **EU compatibility.** Phases follow GSBPM, the transmission is SDMX.

Operations

- **Fast to launch.** One database, three zones, four scheduled loaders.
- **Maintainable by a small team.** No zoo of new technologies.
- **Transparent to audit.** Immutable raw, access logs, reproducible results.
- **Grows in parts.** Scale up only where real demand appears.

The architecture is sized for the task.

A national agency, four sources, well-understood volumes. So the solution stays small and still covers every requirement: sovereignty, confidentiality, integration quality and EU compatibility. A small team can launch it, run it and keep an audit trail that holds up.

Sovereign

PostgreSQL + R

SDC gate is the only exit

Grows in parts

Thank you. Questions welcome.

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